

Program : Diploma in Textile Technology	
Course Code : 6062A	Course Title: Total Quality Management
Semester : 6	Credits: 4
Course Category: Program Elective	
Periods per week: 4 (L:3, T:1, P:0)	Periods per semester: 60

Course Objectives:

- To facilitate the understanding of Quality Management Principles and Process.

Course Prerequisites:

Topic	Course code	Course Title	Semester
Basic concepts of Industrial management		Industrial Management and safety	5

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive level
CO1	Explain the basic concepts of total quality management	15	Understanding
CO2	Summarize the techniques used for continuous process improvement	14	Understanding
CO3	Explain various statistical tools and control charts to measure quality	15	Applying
CO4	Discuss various Management Planning tools	14	Applying
	Series Test	2	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	2					
CO2	3	2					
CO3	3	2					
CO4	3	2					

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive levels
CO1	Explain the basic concepts of total quality management		
M1.01	Discuss the importance of significance of Total Quality Management	8	Understanding,
M1.02	Explain the concept of Quality	7	Understanding
Contents: Basic Concepts: Quality - Definitions - Dimensions of quality - Introduction to TQM - Characteristics - Basic concepts - Pillars - Principles - Obstacles to TQM implementation - Potential benefits of TQM - Quality council - Duties - Responsibilities - Quality statements - Vision - Mission - Quality policy statements - Strategic planning - Customer delight - ISO 9001:2008 Quality Management System requirements and implementation.			
CO2	Summarize the techniques used for continuous process improvement		
M2.01	Identify the principles of TQM	7	Understanding
M2.02	Summarize the techniques used to improve the Quality	7	Understanding
	Series Test - I	1	
Contents: TQM Principles: Input / Output process model - Juran Trilogy - PDCA (Deming Wheel) cycle - 5S Concepts - SEIRI, SEITON, SEISO, SEIKETSU and SHITSUKE - needs and objectives - effective implementation of 5S concepts in an organisation - Housekeeping - Kaizen. Quality Improvement Techniques: Seven tools of quality control (Q-7 tools) - Check sheet - Types of check sheet - Histogram - Cause and effect diagram - Pareto diagram - Stratification Analysis - Scatter diagram-Graph/run charts - Control charts - Construction of above diagrams.			

CO3	Explain various statistical tools and control charts to measure quality		
M3.01	Apply the basic concepts of statistics to quality management and control	8	Applying
M3.02	Illustrate the interpretation of quality control charts	7	Applying
Contents: Statistical Fundamentals: Types of Data - Collection of Data - Classification of Data - Tabular presentation of Data - Graphical representation of a frequency distribution - Mean - Median - Mode - Comparison of measures of central tendency - Introduction to measures of dispersion - Sample - sampling - Normal curve - Sigma - Concept of six sigma - Principles - Process - Problems. Control Charts: Attributes - Control charts - P chart - np chart - c chart - u chart - Construction of above diagrams - Problems - Comparison between variable chart and Attribute chart.			
CO4	Discuss various Management Planning tools		
M4.01	Describe the applications of Advanced Quality tools	7	Applying
M4.02	Explain the need of TQM tools	7	Understanding
	Series Test - II	1	
Contents: Advanced Quality Tools: Affinity diagram - Radar Diagram - Inter Relationship diagram (Inter Relationship diagram) - Tree diagram - Matrix diagram - Decision tree - Arrow diagram - Matrix data analysis diagram - Construction of above diagrams. TQM Tools: Bench marking - Objectives of bench marking - Types - Bench marking process - Benefits of Bench marking - Just In Time(JIT) concepts and its objectives - Total Productive Maintenance(TPM) - Introduction, Objectives of TPM - steps in implementing TPM			

Text /Reference:

T/R	BookTitle/Author
T1	Total Quality Management, Date H.Besterfield, Pearson Education Asia.
R2	Total Quality Management, V.Jayakumar, Lakshmi Publications.(reprint 2005)
R3	Training manual on ISO 9001 : 2000 & TQM, Girdhar J.Gyani, Raj Publishing House, Second Edition 2001
R4	Quality Management, Howard Cuitlow, Tata Mc Graw Hill, 1998
R5	Quality Management - Concepts and Tasks- Narayana.V and Sreenivasan.N.S., New Age International 1996.
R6	Total Quality Management for engineers, Zeiri. Wood Head Publishers. 1991.

Online resources:

Sl.No	Website Link
1	https://nptel.ac.in/courses/110/104/110104080/
2	https://www.vutube.edu.pk/vu-lectures/viewvideo/6506/total-quality-management/total-quality-management-mgmt510-lecture-01
3	https://studylib.net/doc/8794980/lecture-notes-mg1301-%E2%80%93-total-quality-management
4	https://www.youtube.com/watch?v=Pd_uRGy5RKY
5	https://www.slideshare.net/justinsolin/total-quality-management-56112246